UNIVERSITY OF BIRMINGHAM

School of Computer Science

Mathematical and Logical Foundations of Computer Science

Main Summer Examinations 2022

The module was taught differently last year. Only attempt the questions circled in red, not the striked out ones.

Mathematical and Logical Foundations of Computer Science

Question 1

(a) **Numbers and sets**

(i) Compute 4^2 and 4^3 modulo 3. From these results, conjecture a property $P(n)$ of 4^n modulo 3. ~~Prove by induction that $P(n)$ is satisfied for all $n \in \mathbb{N}$.~~ **[3 marks]**

(ii) Let $sqrt : \{x \in \mathbb{N} \mid x \geq 0\} \to \{y \in \mathbb{R} \mid y \geq 0\}$ be the square root function from non-negative integers to non-negative reals, i.e. $sqrt(x) = y$ if and only if $x = y^2$.

Is $sqrt$ injective? surjective? bijective? Justify your answers. **[3 marks]**

(iii) Let `float javaSqrt(int x)` be a Java implementation of this square root function (assume it throws an exception if `x` is negative). Discuss the differences between $sqrt$ and `javaSqrt` when considered as mathematical functions between sets. **[4 marks]**

(b) **Linear algebra**

Consider the three points

$$P_1 = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} \qquad P_2 = \begin{pmatrix} -1 \\ 0 \\ 4 \end{pmatrix} \qquad P_3 = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}$$

(i) Show that these form the corners of an equilateral triangle (i.e. a triangle where all sides have the same length). **[2 marks]**

(ii) The triangle defines a plane M in $\mathbb{R}^3$. Give its parametric representation and its normal form. **[4 marks]**

(iii) Another plane N is given by

$$\begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + q \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + r \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

Find the line of intersection of M and N. **[4 marks]**

Show your working for each part.

Question 2

(a) **Propositional Logic**

Let F be the formula $(A \vee B) \to (B \vee A)$, and
let G be the formula $A \to (B \to (\neg B \vee \neg A)) \to \neg B$.

(i) ~~Provide a constructive Sequent Calculus proof of F.~~ **[4 marks]**

(ii) Provide a constructive Natural Deduction proof of G. **[6 marks]**

(iii) Is G satisfiable? Justify your answer. **[2 marks]**

(b) **Predicate Logic**

Consider the following signature:

- Function symbols: `zero` (arity 0); `succ` (arity 1)
- Predicate symbols: $<$ (arity 2); $\leq$ (arity 2)

We will use infix notation for the binary symbols $<$ and $\leq$. Consider the following formula that captures a property of the above symbols:

- let S_1 be $\forall x.\neg(x < 0)$

For simplicity we write 0 for `zero`, 1 for `succ(zero)`, 2 for `succ(succ(zero))`, etc.

(i) Provide a constructive Natural Deduction proof of:

$$S_1 \rightarrow \neg\exists x.\forall y.x < y$$

[4 marks]

(ii) Provide a model M_1 such that $\models_{M_1} \neg(\forall x.\forall y.x < y \rightarrow x \leq y)$, and a model M_2 such that $\models_{M_2} \forall x.\forall y.x \leq y \rightarrow x < y$ **[4 marks]**